MOHAMMED VI
FACULTY OF MEDICINE
UM6SS - CASABLANCA

Nausea - Vomiting

Pr AFAFE TAIYMI

Module: Digestive Diseases

Academic year: 2026-2027

MOHAMMED VI UNIVERSITY
OF SCIENCES AND HEALTH
UM6SS

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Learning objectives

Be able to define nausea and vomiting.
Understand the semiological characteristics
Identify the various etiologies of vomiting.
Select appropriate diagnostic tests to identify underlying conditions contributing to vomiting.

LAY OUT

I. Definition
II. Pathophysiology
III. Diagnostic approach
IV. Differential diagnosis
V. Aetiologies
VI. Complications
VII. Conclusion

1 Definition

Nausea : is defined as an unpleasant, subjective sensation often associated with the urge to vomit, typically originating from the gastrointestinal tract or the central nervous system.



1 Definition

Vomiting : or emesis , is the forceful expulsion of stomach contents through the mouth, usually preceded by nausea. It involves an active mechanism of violent cyclic contractions of the abdominal muscles, diaphragm, and respiratory muscles.



1 Definition

- One of the most common gastrointestinal complaints

- **Several Etiologies**

- Nausea and vomiting are symptoms that can signal a wide range of medical conditions, from minor gastrointestinal disases to more serious neurological or systemic issues.



2 Pathophysiology

Vomiting can be triggered by various stimuli :

- Gastrointestinal irritation
- Vestibular system disturbances
- Higher cortical centers stimulation
- Chemoreceptor trigger zone (CTZ) stimulation by drugs or toxins

↓

Vomiting center Activation

↓

Abdominal vagal afferents Activation

↓

Strong contractions of the diaphragm and abdominal muscles

Relaxation of the lower esophageal sphincter

↓

Forceful ejection of stomach contents through the mouth.

→ Vomiting

1. Patient History :

Age
Medical and Surgical History: Especially digestive
Medication Use: Type, dosage, and date of the last
Exposure to Toxic Substances: Any history of toxic substance intake
History of Recent travel
Family or Close Contacts: Similar cases or symptoms in close contacts or family members
Menstrual History and Contraception: Date of last menstrual period and use of contraception

2. Semiological characteristics of vomiting :

Onset and Duration: Acute or chronic or recurrent

Circumstances of occurrence : Spontaneous or induced

Frequency: Daily, Several times a day, Occasional

Content of Vomit: Food particles, bile, blood.

Volume: Mild or abundant

Triggers and Relieving Factors: Diet, changes in head position

2. Semiological characteristics of vomiting :

Pattern and timing :

Morning Vomiting : Often seen in pregnant women or individuals with alcohol dependence.

Postprandial Vomiting:

Early Postprandial Vomiting: May suggest gastric emptying or gastroparesis.

Late Postprandial Vomiting: May suggest a duodenal or pyloric obstruction.

Vomiting During Meals: Psychogenic cause

Impact on Patient: Effect on appetite, hydration status, weight loss, and overall physical condition.

2. Semiological characteristics of vomiting :

Content of vomit significance :	
Clear Fluid	Suggests acid reflux or regurgitation,
Undigested Food	Suggests gastric emptying disorders, such as gastroparesis or gastric outlet obstruction.
Bile-Stained (Green or Yellow)	Suggests bile reflux.
Blood-Stained (Hematemesis)	Suggests active bleeding, from the upper gastrointestinal tract
Fecal Odor or Feculent Vomit	Strongly suggests severe bowel obstruction

3. Associated Symptoms:

Nausea,
Abdominal pain,
Diarrhea,
Myalgias
Fever,
Extra digestive symptoms : headache, Photophobia (meningeal syndrome) , Altered consciousness or any other relevant symptoms.

4. Physical examination :

General assessment :

Signs of dehydration : Dry mucous membranes, reduced skin turgor , sunken eyes.

Check vital signs : Heart rate , blood pressure and respiratory rate.

Temperature





4. Physical examination :
Abdominal Examination:

Inspection: Abdominal Distention, scars from previous surgeries.

Palpation: Tenderness, masses.

Auscultation: Bowel sounds.

Hyperactive bowel sounds → May indicate obstruction,

Absent sounds may suggest → Ileus.

Percussion: Assess for fluid (ascites) or areas of dullness that may suggest an abdominal mass .

Alarm Symptoms

Persistent vomiting / inability to maintain oral intake

Severe or progressive abdominal pain

Hematemesis

Fever

Neurological symptoms: severe headache, altered consciousness,

Signs of dehydration

Unintentional weight loss

Bilious or feculent vomiting → suggests intestinal obstruction

Elderly patient

Immunocompromised state

4. Physical examination :

Neurological Examination:

Headache, papilledema.

→Signs of raised intracranial pressure

Altered mental status, dizziness, or nystagmus

→ Central nervous system causes of vomiting

Cardiovascular Examination:

Any signs of heart failure if cardiac causes are suspected

Lymphadenopathies Examination :

→Suggests infections or malignancy

Additional Examination Based on History:

5. Paraclinical tests :

Aims :

To provide an initial workup to assess the possible etiologies of vomiting,
To detect complications,
To guide further diagnostic and therapeutic decisions.

The choice of specific tests depends on the clinical presentation and suspected etiology.

5. Paraclinical tests :

The initial recommended paraclinical tests are :

Lab tests :

Complete Blood Count (CBC)
Serum Electrolytes : Sodium, potassium, chloride, bicarbonate
Renal Function Tests.
Glucose Levels: rule out diabetic ketoacidosis (DKA)
Pregnancy Test (hCG) : In women of reproductive age
Thyroid Function Tests: rule out hyperthyroidism if clinical suspicion arises
Urinalysis : Ketones

5. Paraclinical tests :

Endoscopy : Upper GI endoscopy

Aims :

Combines diagnostic evaluation with the option for immediate biopsy.

Indication :

Suspected mucosal lesion,
Obstruction,
Complications : upper GI bleeding , Esophagitis or gastritis

Endoscopic findings :

Ulcers, Erosions, Masses
Signs of chronic GERD or Barrett's esophagus:

3 Diagnostic approach



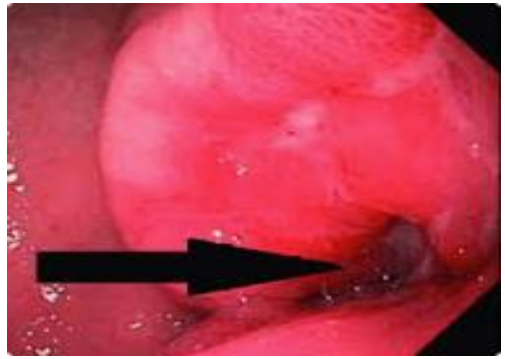
Erosions



Gastric ulcer



Gastric cancer



Pylorus obstruction

5. Paraclinical tests :

Imaging

Ultrasound: evaluate the gallbladder, liver, pancreas, and kidneys. It can also help identify conditions like cholecystitis or appendicitis.

CT Scan (If indicated): Provides detailed imaging in cases where obstruction, pancreatitis, or other intra-abdominal pathology is suspected.

5. Paraclinical tests :

Other paraclinical tests :

Electrocardiogram (ECG):
If electrolyte imbalance is suspected, particularly hypokalemia, which can affect heart rhythm.

Cerebral CT-Scan

Endocrine or toxicological tests if indicated

4 Differential diagnosis

Gastroesophageal regurgitation (GER):

The backward flow of stomach contents into the esophagus without nausea or the feeling of vomiting , often associated with acid reflux or gastroesophageal reflux disease (GERD).

Rumination Syndrome :

Repeated regurgitation of food or liquid from the stomach into the mouth, without nausea or the feeling of vomiting.

5 Etiologies

1. Acute Vomiting :

Gastrointestinal Causes:

Gastroenteritis	Surgical acute abdomen causes	Non surgical acute abdomen causes
Viral, bacterial, or parasitic Food poisoning	• Peritonitis • Appendicitis • Cholecystitis • Intestinal obstruction	Acute pancreatitis Biliary colic Peptic ulcer disease Acute gastritis (often due to NSAIDs)

1. Acute Vomiting :
Non-gastrointestinal causes :

Non-gastrointestinal causes :	
Metabolic and Endocrine Causes	Diabetic ketoacidosis Adrenal crisis Hyperthyroidism Hypercalcemia
Neurological Causes	Increased intracranial pressure : Head injury, stroke, or brain tumor Meningitis Migraine headaches Vestibular disorders
Cardiovascular Causes	Myocardial Infarction Aortic Dissection
Infectious Causes	Sepsis Pyelonephritis

1. Acute Vomiting :
Non-gastrointestinal causes :

Non-gastrointestinal causes :	
Pregnancy	Hyperemesis Gravidarum: Severe and persistent vomiting during pregnancy
Electrolyte Imbalances	Hypokalemia Hyponatremia
Toxic Causes	Drug toxicity : Chemotherapy, antibiotics, opioids Alcohol intoxication Toxic ingestion : Heavy metals, toxins
Psychogenic Causes:	Anxiety or stress-related vomiting Eating disorders : Bulimia

2. Chronic vomiting :

Gastro-intestinal causes :

Pyloroduodenal stenosis
Gastroesophageal reflux disease (GERD)
Gastroparesis : often associated with diabetes
Chronic pancreatitis
Irritable bowel syndrome (IBS)
Chronic intestinal pseudo-obstruction
Celiac disease
Gastric cancer

2. Chronic vomiting :
Non Gastro-intestinal causes :

Non-gastrointestinal causes :	
Metabolic and Endocrine Causes	Chronic renal failure : Uremic syndrome Hyperthyroidism Adrenal Insufficiency (Addison's Disease)
Neurological Causes:	Chronic migraine Intracranial tumors Chronic increased intracranial pressure
Toxic and Drug-Induced Causes	Long-term medication side effects (e.g., NSAIDs, opioids, antibiotics) Chronic alcohol use
Psychiatric Causes	Functional vomiting Eating disorders : Binge-purge behavior Psychogenic vomiting : often in response to stress or trauma

Hydro-electrolytic disorders

Electrolyte Imbalances :

- Hypokalemia
- Hyponatremia
- Metabolic alkalosis.



Dehydration

Mechanical complication

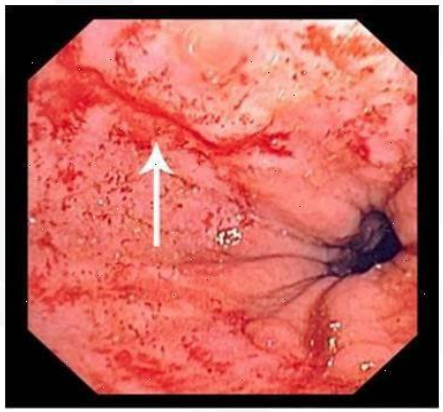
Esophageal Tears (Mallory-Weiss Syndrome) :

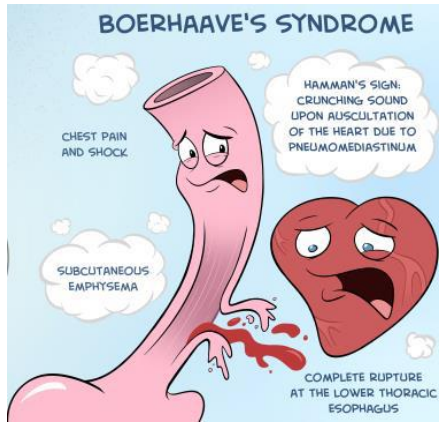
Forceful vomiting can cause tears in the mucosa at the junction of the stomach and esophagus, leading to bleeding.

Esophageal rupture (Boerhaave syndrome) :

A life-threatening condition.

Caused by a sudden increase in intra-abdominal pressure from forceful vomiting → Tear in the esophageal mucosa and leakage of contents into the mediastinum.



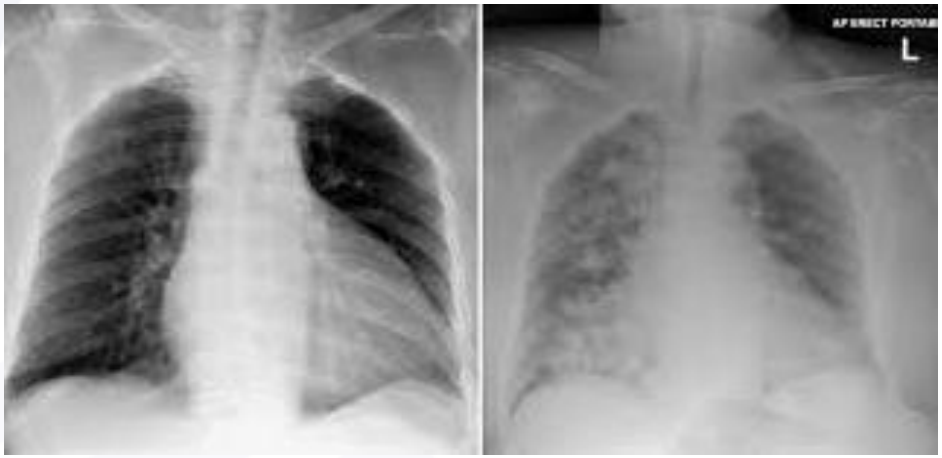


Infectious complications

Aspiration Pneumonia :

Result in the inhalation of stomach contents into the lungs, leading to aspiration pneumonia

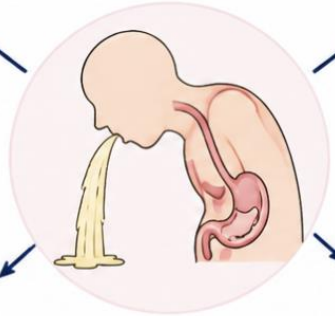
Causes fever, cough, and difficulty breathing.



VOMITING
Clinical approach

1. HISTORY
- Age
- Onset (acute < 2 weeks / chronic > 3 weeks)
- Circumstances (spontaneous or induced)
- Frequency (occasional, daily, ...)
- Relation to meals
- Triggers (diet, head position, motion, odors, drugs...)
- Medical and surgical history
- Medications and toxins
- Travel history
- Menstrual history / pregnancy
- Similar cases in close contacts

2. CHARACTERISTICS
- **Pattern:** acute / chronic / recurrent
- **Timing:** morning, postprandial (early/late), during meals
- **Volume:** mild / moderate / abundant
- **Impact:** appetite, hydration, weight loss



3. CONTENT OF VOMIT

Clear fluid	Acid reflux / regurgtiation
Undigested food	Gastric emptying disorder (gastroparesis)
Bile (green/yellow)	Bile reflux / duodenal obstruction
Blood (hematemesis)	Upper GI bleeding
Feculent (fecal odor)	Bowel obstruction

4. ASSOCIATED SYMPTOMS
- Nausea
- Abdominal pain
- Bloating
- Altered bowel habits (diarrhea/constipation)
- Fever
- Headache, dizziness
- Weight loss
- Dehydration signs
- Others (e.g. myalgia, arthralgia...)

5. ALARM FEATURES
- Persistent vomiting
- Severe abdominal pain
- Hematemesis
- Fever
- Neurological symptoms (headache, altered consciousness)
- Signs of dehydration
- Weight loss
- Elderly patient
- Immunocompromised state

6. ORIENTATION
(main etiologic groups)

GASTROINTESTINAL
- Gastroenteritis
- Gastric outlet obstruction
- Peptic ulcer disease
- Gastroparesis
- Pancreatitis
- Biliary disease
- Bowel obstruction

CENTRAL NERVOUS SYSTEM
- Increased intracranial pressure (tumor, hemorrhage)
- Migraine
- Vertigo (vestibular)
- Meningitis / encephalitis

METABOLIC / SYSTEMIC
- Uremia
- Diabetic ketoacidosis
- Electrolyte disorders (Na^+, K^+, Ca^{2+})
- Hepatic failure
- Endocrine (pregnancy, adrenal insufficiency, ...)

DRUGS / TOXINS
- Chemotherapy
- Opioids, antiboctics, NSAIDs, digoxin, ...
- Alcohol
- Toxins

PSYCHOGENIC
- Psychogenic vomiting
- Eating disorders
- Rumination syndrome

7 Conclusion

Vomiting is a common clinical symptom with a broad range of potential causes

A detailed history and clinical examination are essential for determining the underlying etiology.

While digestive disorders are often the primary focus, extra gastrointestinal causes should not be overlooked.